

SECTION A

1 (**short**) The hinged, rectangular gate shown in Fig. 1 has a width w into the page, and retains water of density ρ that reaches a height h .

(a) Find an expression for the total hydrostatic force acting on the gate. [4]

(b) Find an expression for the reaction R necessary to keep the gate closed as a function of the water height h . For values of h between 0 and $4L$, sketch $R(h)$. [6]

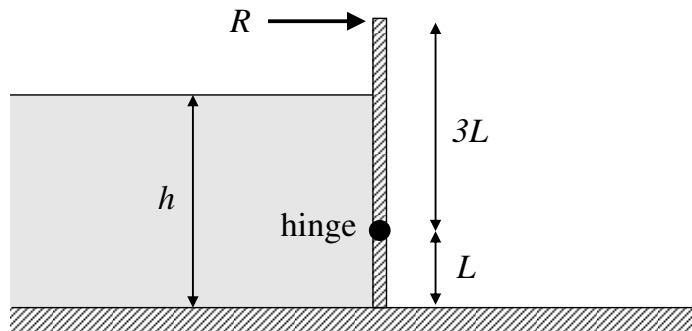


Fig. 1